EDA and Data Cleaning by Column

Six encounter-level CSV splits | 101,766 records per split | Final joined dataset: 101,766 rows and 56 columns

How to read this report

Each source feature is listed once per split. EDA means the feature was inspected and, where appropriate, summarized or plotted. DROP means the source feature was removed from the cleaned output. The last column lists the resulting field names; blank cells indicate a dropped field.

Cleaning decisions

The literal '?' was read as missing. Diagnosis codes and other category codes were kept as text/codes. Missing diagnosis, payer, and race entries were labeled Unknown for analysis while a *_missing indicator preserves the missingness. Missing glucose/A1C test results were labeled Not measured with indicators. Age intervals were split into lower and upper bounds. Weight was dropped at 96.86% missingness. Constant citoglipton and examide were dropped. No rows were removed; no duplicate rows or encounter_id keys were found.

Near-constant medication features were retained because their rare non-No values still vary and may carry predictive information. The diagnosis code meanings are not expanded in the cleaned file; no code descriptions are implied by this report.

Dataset 1 columns

Feature / column name	EDA or DROP	Description	Inference / output
encounter_id	EDA	Checked for missing values and duplicates. Confirmed unique across all 101,766 encounters; used to align and join all six files.	encounter_id (preserved as join key)
num_medications	EDA	Checked missingness, unique values, summary statistics, frequency/distribution, and box plot.	num_medications
metformin-pioglitazone	EDA	Checked category counts and plotted category frequencies. Rare medication status retained because it is not constant.	metformin-pioglitazone
diabetesMed	EDA	Checked category counts and plotted the Yes/No frequency distribution.	diabetesMed
metformin-rosiglitazone	EDA	Checked category counts and plotted category frequencies. Rare non-No values retained.	metformin-rosiglitazone
diag_1	EDA	Diagnosis code kept as text. Checked missingness, code frequencies, and top diagnosis codes; source '?' was treated as missing and labeled Unknown for analysis.	diag_1 diag_1_missing
glimepiride	EDA	Checked medication-status counts and plotted category frequencies.	glimepiride
chlorpropamide	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	chlorpropamide
nateglinide	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	nateglinide
tolbutamide	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	tolbutamide

Dataset 2 columns

Feature / column name	EDA or DROP	Description	Inference / output
encounter_id	EDA	Checked for missing values and duplicates. Confirmed unique and aligned with the other splits.	encounter_id (preserved as join key)
diag_3	EDA	Diagnosis code kept as text. Checked missingness, code frequencies, and top diagnosis codes; source '?' was treated as missing and labeled Unknown for analysis.	diag_3 diag_3_missing
num_procedures	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, box plot, and numeric correlation with other measures in this split.	num_procedures
weight	DROP	Dropped because 98,569 of 101,766 values were missing (96.86%); too sparse for reliable use in this split.	
acarbose	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	acarbose
time_in_hospital	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, box plot, and correlation with other numeric measures.	time_in_hospital
age	DROP	Source age band was expanded into numeric lower and upper bounds, then the original interval text was removed.	age_min age_max
discharge_disposition_id	EDA	Checked missingness, category counts, unique codes, and category-frequency plot. IDs remain codes; no unsupported labels were inferred.	discharge_disposition_id
insulin	EDA	Checked medication-status counts and plotted category frequencies.	insulin

Dataset 3 columns

Feature / column name	EDA or DROP	Description	Inference / output
encounter_id	EDA	Checked for missing values and duplicates. Confirmed unique and aligned with the other splits.	encounter_id (preserved as join key)
glyburide-metformin	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	glyburide-metformin
change	EDA	Checked category counts and plotted whether medication changes were recorded.	change
number_emergency	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, box plot, and correlation with inpatient visits.	number_emergency
payer_code	EDA	Checked missingness and payer-code frequencies. Source '?' was treated as missing and labeled Unknown for analysis.	payer_code payer_code_missing
number_inpatient	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, box plot, and correlation with emergency visits.	number_inpatient
repaglinide	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	repaglinide
rosiglitazone	EDA	Checked medication-status counts and plotted category frequencies.	rosiglitazone
glyburide	EDA	Checked medication-status counts and plotted category frequencies.	glyburide

Dataset 4 columns

Feature / column name	EDA or DROP	Description	Inference / output
encounter_id	EDA	Checked for missing values and duplicates. Confirmed unique and aligned with the other splits.	encounter_id (preserved as join key)
patient_nbr	EDA	Checked identifier completeness and distinct-patient count. Preserved because many encounters can belong to the same patient.	patient_nbr
glimepiride-pioglitazone	EDA	Checked medication-status counts and plotted category frequencies. Rare status retained.	glimepiride-pioglitazone
pioglitazone	EDA	Checked medication-status counts and plotted category frequencies.	pioglitazone
admission_type_id	EDA	Checked missingness, unique codes, category counts, and category-frequency plot. Kept as an ID code.	admission_type_id
glipizide	EDA	Checked medication-status counts and plotted category frequencies.	glipizide
num_lab_procedures	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, and box plot.	num_lab_procedures
troglitazone	EDA	Checked medication-status counts and plotted category frequencies. Rare status retained.	troglitazone
race	EDA	Checked missingness and category distribution. Source '?' was treated as missing and labeled Unknown for analysis.	race race_missing

Dataset 5 columns

Feature / column name	EDA or DROP	Description	Inference / output
encounter_id	EDA	Checked for missing values and duplicates. Confirmed unique and aligned with the other splits.	encounter_id (preserved as join key)
max_glu_serum	EDA	Checked category distribution and source missingness. Missing means no recorded result; labeled Not measured and retained a missingness flag.	max_glu_serum max_glu_serum_missing
gender	EDA	Checked category counts and plotted frequencies. Normalized Unknown/Invalid to Unknown.	gender
tolazamide	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	tolazamide
miglitol	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	miglitol
metformin	EDA	Checked medication-status counts and plotted category frequencies.	metformin
citoglipton	DROP	Dropped because every encounter had the same value (No); a constant feature has no variation for analysis or prediction.	
medical_specialty	EDA	Checked missingness and specialty frequencies. Source '?' was treated as missing and labeled Unknown for analysis.	medical_specialty medical_specialty_missing
A1Cresult	EDA	Checked category distribution and source missingness. Missing means no recorded result; labeled Not measured and retained a missingness flag.	A1Cresult A1Cresult_missing

Dataset 6 columns

Feature / column name	EDA or DROP	Description	Inference / output
encounter_id	EDA	Checked for missing values and duplicates. Confirmed unique and aligned with the other splits.	encounter_id (preserved as join key)
diag_2	EDA	Diagnosis code kept as text. Checked missingness, code frequencies, and top diagnosis codes; source '?' was treated as missing and labeled Unknown for analysis.	diag_2 diag_2_missing
readmitted	EDA	Checked outcome category counts and plotted readmission frequencies; compared diagnosis counts and prior outpatient visits by readmission group.	readmitted
number_diagnoses	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, box plot, numeric correlation, and grouped summaries by readmission outcome.	number_diagnoses
admission_source_id	EDA	Checked missingness, unique codes, category counts, and category-frequency plot. Kept as an ID code.	admission_source_id
glipizide-metformin	EDA	Checked medication-status counts and plotted category frequencies. Rare statuses retained.	glipizide-metformin
number_outpatient	EDA	Checked missingness, unique values, descriptive statistics, frequency/distribution, box plot, correlation, and grouped summaries by readmission outcome.	number_outpatient
acetohexamide	EDA	Checked medication-status counts and plotted category frequencies. Rare status retained because it is not constant.	acetohexamide
examide	DROP	Dropped because every encounter had the same value (No); a constant feature has no variation for analysis or prediction.	

Output files

Cleaned split files and EDA notebooks are in the project. The joined output is EDA_Dataset/EDA_Full_Dataset.csv. The join uses encounter_id with one-to-one validation.